WIC ANALYSIS No Code

An RDD Design

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Special Supplemental Nutrition Program for Women, Infants, and Children

PA 528 Dr. Siciliano

Introduction

The motivation for evaluating the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC) stems from its critical role in addressing the nutritional needs of vulnerable populations. Established in 1972, WIC serves low-income mothers, infants, and young children by providing access to healthy food, nutrition education, and healthcare referrals. As of 2024, WIC benefits nearly 53% of all infants born in the United States. Evaluating this program is vital because it addresses issues such as food insecurity, low birth weights, and developmental challenges. Understanding its impact informs policymakers and ensures continued investment in evidence-based solutions to promote public health.

Program Overview

WIC was launched in 1972 as a pilot program to combat the nutritional deficiencies observed among low-income families, particularly women and children. It became a permanent federal program in 1975, amid growing societal concerns about equality and public health. The program's primary focus is on early nutritional intervention, which research shows can mitigate medical and developmental issues later in life.

Initially serving 88,000 participants per month, WIC has expanded significantly over the decades, reaching 7.2 million monthly participants by 2000. Its benefits include supplemental foods, nutrition education, and healthcare referrals. The program has shown measurable impacts, such as reducing rates of low birth weights, premature births, and child anemia, as well as improving overall development. Its continued success is attributed to strong congressional backing and data demonstrating its cost-effectiveness in preventing long-term health complications.

Program Theory and Implementation

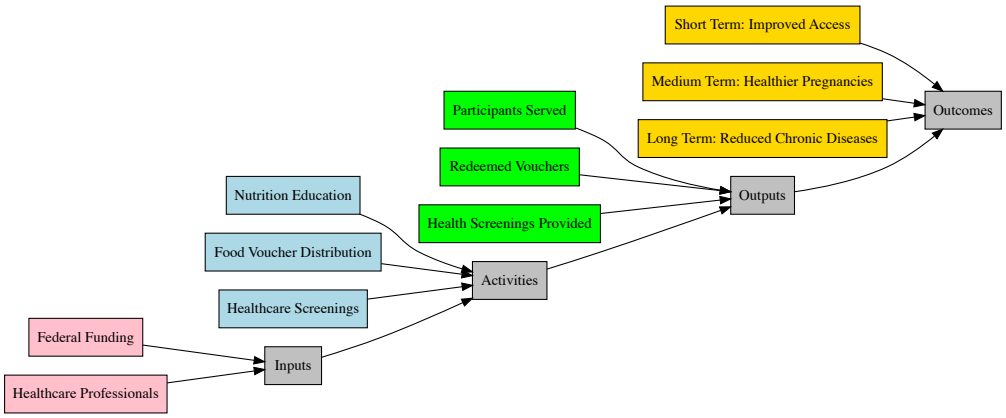
WIC operates on the premise that addressing nutritional deficiencies among pregnant women and young children leads to improved health outcomes. Grounded in public health research, the program targets key developmental stages with interventions like food vouchers, health screenings, and nutrition education. Studies have shown that these activities promote healthier pregnancies, higher birth weights, and cognitive development. The program also encourages breastfeeding, emphasizing its benefits for immunity and mother-child bonding. These elements reflect a well-articulated theory of change based on established health and nutrition science.

Logic Model

Inputs: Federal funding, healthcare professionals, and local clinics.
Activities: Nutrition education, food voucher distribution, health screenings, and referrals.
Outputs: Number of participants served, vouchers redeemed, and health screenings conducted.
Outcomes:

- **Short-term:** Improved nutrition and access to healthcare.
- **Intermediate:** Healthier pregnancies and reduced child anemia.
- **Long-term:** Reduced chronic disease prevalence and improved cognitive outcomes.

The logic model visualizes how resources are transformed into tangible activities and outcomes. The linkages between each stage ensure accountability and efficacy.


Short Term: Improved Access
Medium Term: Healthier Pregnancies
Long Term: Reduced Chronic Diseases
Outcomes
Participants Served
Redeemed Vouchers
Health Screenings Provided
Outputs
Nutrition Education
Food Voucher Distribution
Healthcare Screenings
Activities
Federal Funding
Healthcare Professionals
Inputs

Causation

Main Outcome

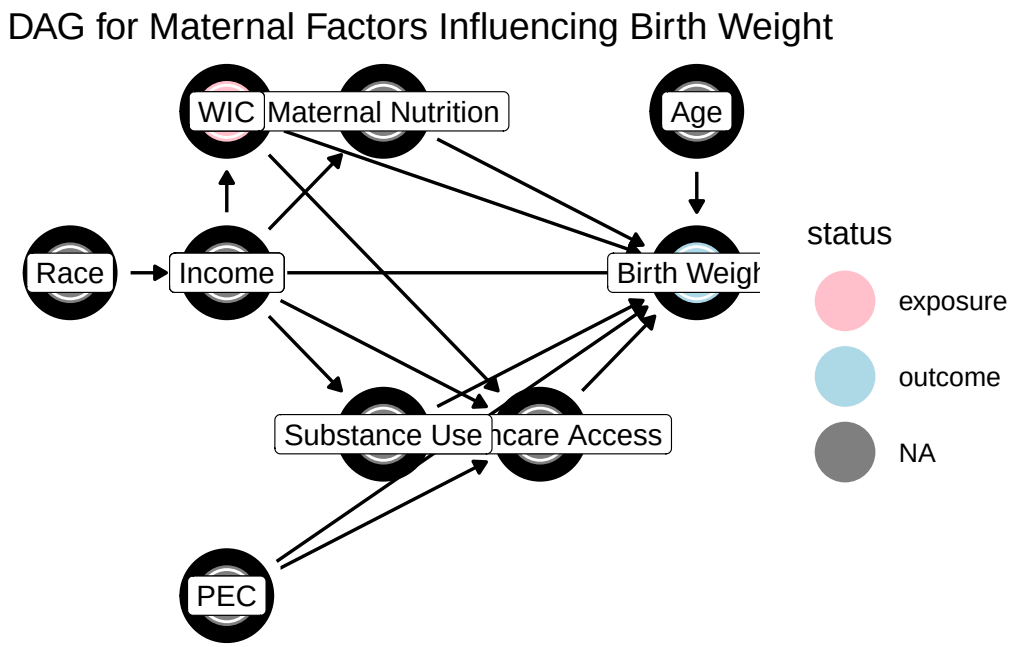
The primary outcome selected is a reduction in low birth weights, as it is directly linked to the program's nutritional interventions and has profound implications for public health.

Causal Theory

The causal pathway hypothesizes that WIC participation improves maternal nutrition and healthcare access, influencing birth weight outcomes. WIC's supplemental foods, nutrition education, and healthcare referrals target the main determinants of healthy pregnancies. Improved maternal nutrition reduces risks like low birth weight, while better healthcare access assures timely prenatal interventions.

The model includes socioeconomic status (SES), maternal education, and pre-existing health conditions as control variables to block alternative causal pathways. SES affects access to healthcare and other resources, maternal education influences health literacy and decision-making, and health conditions independently impact birth outcomes. By including these factors, WIC's impact is isolated while confounding is minimized.

The Directed Acyclic Graph (DAG) illustrates relationships with nodes representing variables and arrows indicating causal links. SES, education, and health conditions influence maternal nutrition and healthcare, impacting birth weight directly. The model ensures a robust evaluation of WIC's effects on maternal/child health by controlling for external factor



Hypotheses

Participation in WIC will reduce the incidence of low birth weights. The research question asks: Does WIC participation causally improve birth outcomes among low-income mothers?

Data and Methods

Identification Strategy

Regression Discontinuity (RD) will be performed analyzing participants near the income eligibility threshold, ensuring local comparability and minimizing bias. Threats to validity include manipulation of income data near the threshold and unobserved time-varying factors

Given the nature of our program, income eligibility is the threshold for cutoff, which allows for causal inference by comparing individuals above and below the threshold. By focusing on a narrow bandwidth of income near the cutoff, RDD ensures local comparability, minimizes selection bias, and enhances internal validity.

This framework accounts for potential selection bias and endogeneity by exposing quasi-random assignment of program eligibility near the threshold. It assumes individuals on either side of the cutoff are similar in all aspects except for program participation, enabling isolation of the causal effects and attributing differences in birth weight outcomes to the WIC involvement. Administrative data will further strengthen the analysis, providing accurate and consistent measures of income, birth weight, and other essential variables, reducing the risk of reporting bias.

Despite its strengths, the study faces threats to internal and external validity, as internal manipulation of income data near the threshold could bias the results if individuals deliberately misreport their income to qualify for WIC. To mitigate this, density plots will be used to check for robustness and irregular income distribution near the cutoff. Also, unobserved time-varying factors may influence outcomes, such as economic conditions or policy changes. These factors will be controlled by including covariates to capture temporal trends.

Externally, RDD's localized nature limits generalizability as its findings primarily apply to individuals near the eligibility threshold. Broader populations with significantly different socioeconomic profiles can experience different effects. Nevertheless, the RDD approach provides robust causal estimates and insights into WICs' efficiency in addressing nutritional deficiencies and improving maternal/infant health outcomes.

Data

The analysis will use administrative and synthetic data to assess WIC's impact on birthweight. To ensure accuracy, administrative data will be collected from WIC clinics and health departments, including birth weights, maternal nutrition, and SES metrics. While HIPAA laws limit access, partnerships with local facilities ensure feasibility.

Additionally, a synthetic dataset of 1,000 observations will simulate variables such as income, WIC participation, SES, and birth weight. The dataset intends to account for confounders such as maternal education and pre-existing conditions while having mediators such as nutrition. This combined approach ensures a robust method like Regression Discontinuity Design (RDD) can ground its findings in real-world applicability.

Synthetic Analysis

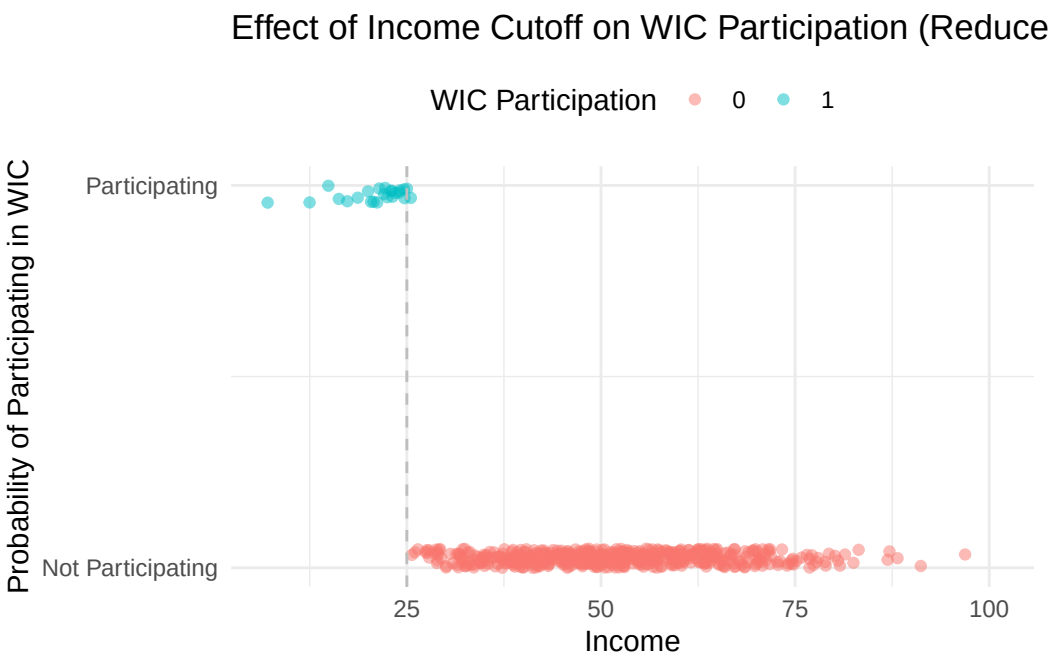
Setup

Generate Data

Applying RDD

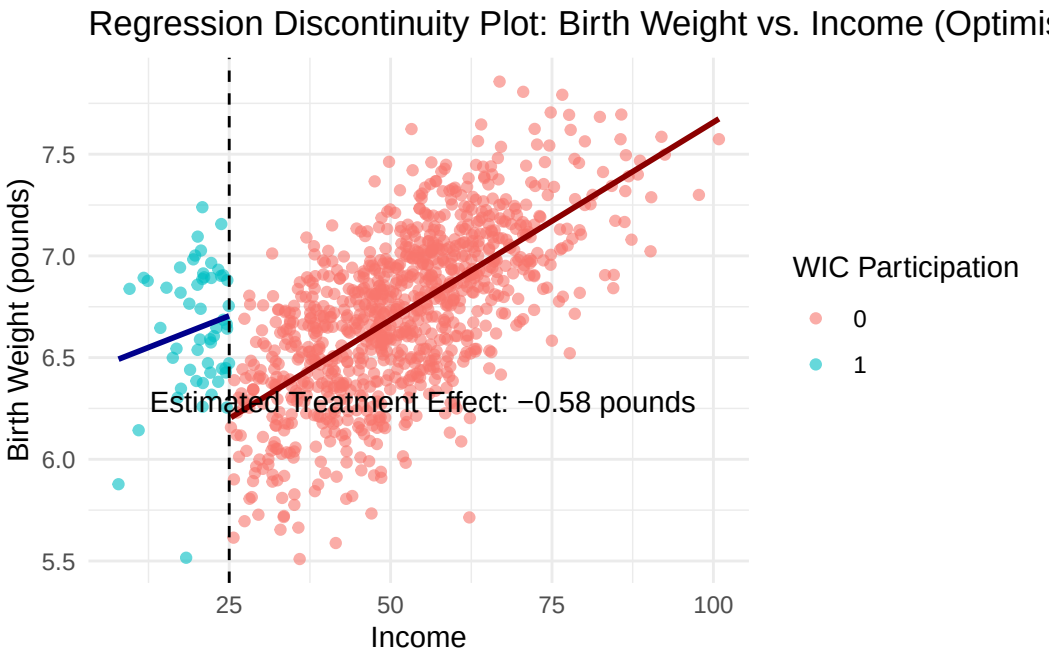
```
Sharp RD estimates using local polynomial regression.

Number of Obs.                 1000
BW type                       mserd
Kernel                   Triangular
VCE method                       NN

Number of Obs.                   55          945
Eff. Number of Obs.              33           38
Order est. (p)                    1            1
Order bias  (q)                   2            2
BW est. (h)                   4.814        4.814
BW bias (b)                   7.981        7.981
rho (h/b)                     0.603        0.603
Unique Obs.                      55          945

=============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
  Conventional  -262.730    66.213    -3.968     0.000  [-392.505 , -132.955]
        Robust         -         -    -3.380     0.001  [-425.436 , -113.138]
=============================================================================
```

Visualizing Cutoff For RDD



In this graph demonstrating the cutoff mark for participation (1) and non-participation (0), the dotted line signifies one's eligibility for the WIC program. For individuals with income levels below that line, participation in the program is nearly universal, as shown by the cluster of blue points. For those above the line, participation is nonexistent, demonstrating a sharp discontinuity cutoff for the program based on income levels. The sharp change in participation at the cutoff supports the validity of using RDD to estimate the causal effect of WIC participation on the outcomes of interest (birth weight or premature births) by comparing individuals just below and above the income threshold.

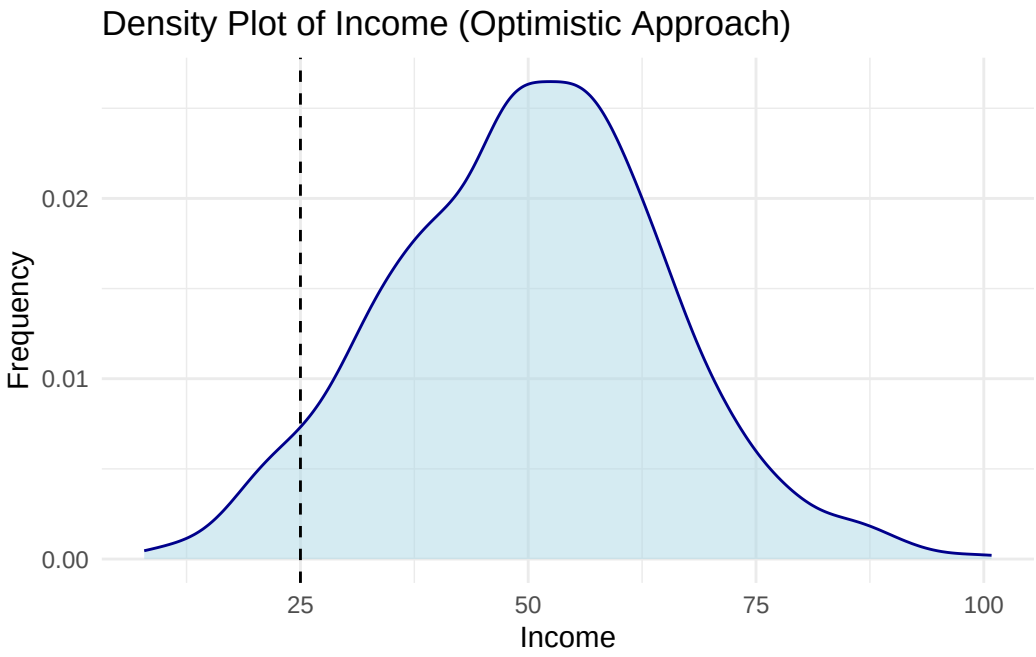
Visualizing Treatment Effect : Optimistic Model



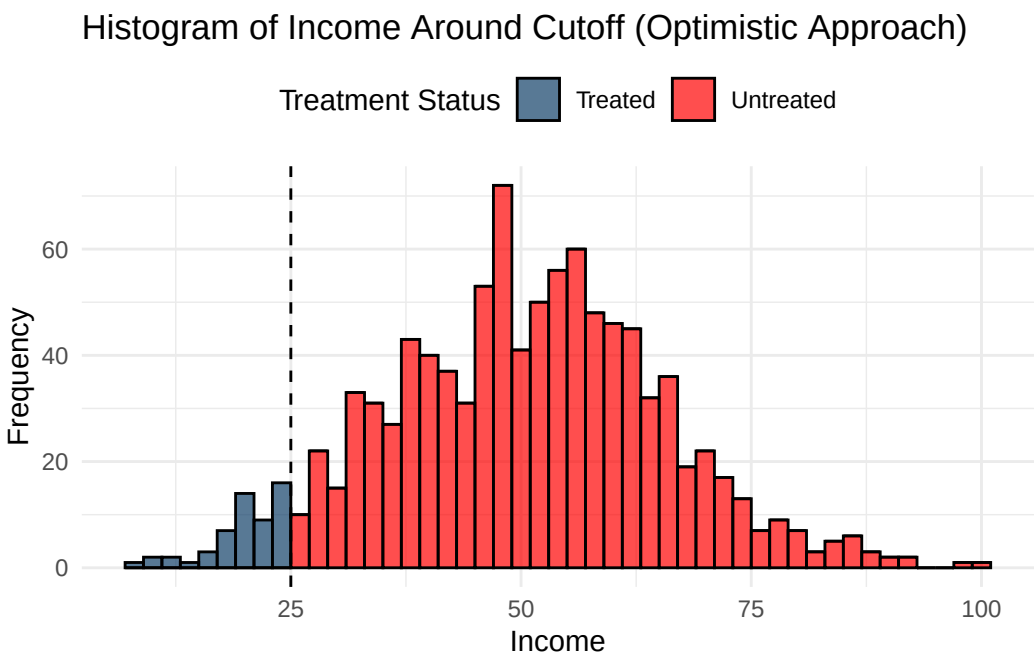
This graph shows a regression discontinuity plot that explores the relationship between income, WIC participation, and birth weight, where the dotted line represents the income cutoff for program participation, and how participation impacts the birth weight for children/families who do or do not partake in the program. The regression line for participants (blue) and non-participants (red) shows a slight positive slope, indicating that birth weight generally increases with income in both groups. However, the sharp drop at the income cutoff highlights the distinct impact of WIC participation. The estimated treatment effect is labeled as 0.58 pounds, meaning that WIC participation is associated with an increase in birth weight by 0.58 pounds compared to those who do not participate.

Include Robustness Checks

First, we check for manipulation using density plots. Initial approach makes use of a line graph and then use the histogram approach.



This graph utilizes the data that shows the distribution of income for individuals who are eligible for the WIC program and those who are not, using the optimistically created data. The dashed line indicates where the treatment (WIC participation) is determined based on income. The distribution's continuity around this point is crucial for ensuring the validity of the regression discontinuity design. If there is no discontinuity in the density at the threshold, it supports the assumption that income assignment near the threshold is as good as random. The smooth curve near the dashed line suggests that there is no evident manipulation of income to qualify or disqualify for WIC. This strengthens the causal interpretation of the estimated treatment effect from the previous RD analysis.



Using the optimistically created data, a histogram was created to see the distribution of income between individuals who participate in the WIC program and those who were ineligible to do as their incomes were greater than the cutoff. There is a sharp change in the number of individuals on either side of the cutoff, reflecting the income eligibility criteria for WIC and indicates that households are being clearly sorted into treated and untreated groups based on their income. If the WIC program is effective, we would expect to see significant differences in outcomes (better birth weights or improved nutrition) between the treated and untreated groups, particularly for those just around the income cutoff. The shape of the histogram might also suggest areas for policy adjustments—for instance, if there are many households just above the cutoff, they may benefit from expanded eligibility thresholds.

```
Sharp RD estimates using local polynomial regression.

Number of Obs.                 1000
BW type                      Manual
Kernel                   Triangular
VCE method                       NN

Number of Obs.                   55          945
Eff. Number of Obs.              49          111
Order est. (p)                    1            1
Order bias  (q)                   2            2
BW est. (h)                  10.000       10.000
```

```
BW bias (b)                  10.000       10.000
rho (h/b)                     1.000        1.000
Unique Obs.                      55          945

==============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
==============================================================================
  Conventional  -228.844    43.898    -5.213     0.000  [-314.883 , -142.805]
        Robust         -         -    -3.705     0.000  [-377.841 , -116.399]
==============================================================================


Sharp RD estimates using local polynomial regression.

Number of Obs.                 1000
BW type                      Manual
Kernel                   Triangular
VCE method                       NN

Number of Obs.                   55          945
Eff. Number of Obs.              36           39
Order est. (p)                    1            1
Order bias  (q)                   2            2
BW est. (h)                   5.000        5.000
BW bias (b)                   5.000        5.000
rho (h/b)                     1.000        1.000
Unique Obs.                      55          945

==============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
==============================================================================
  Conventional  -257.913    64.722    -3.985     0.000  [-384.766 , -131.060]
        Robust         -         -    -3.078     0.002  [-531.723 , -118.029]
==============================================================================
```

Now for added rigor and fun, we repeat the process with our pessimistic data simulation.

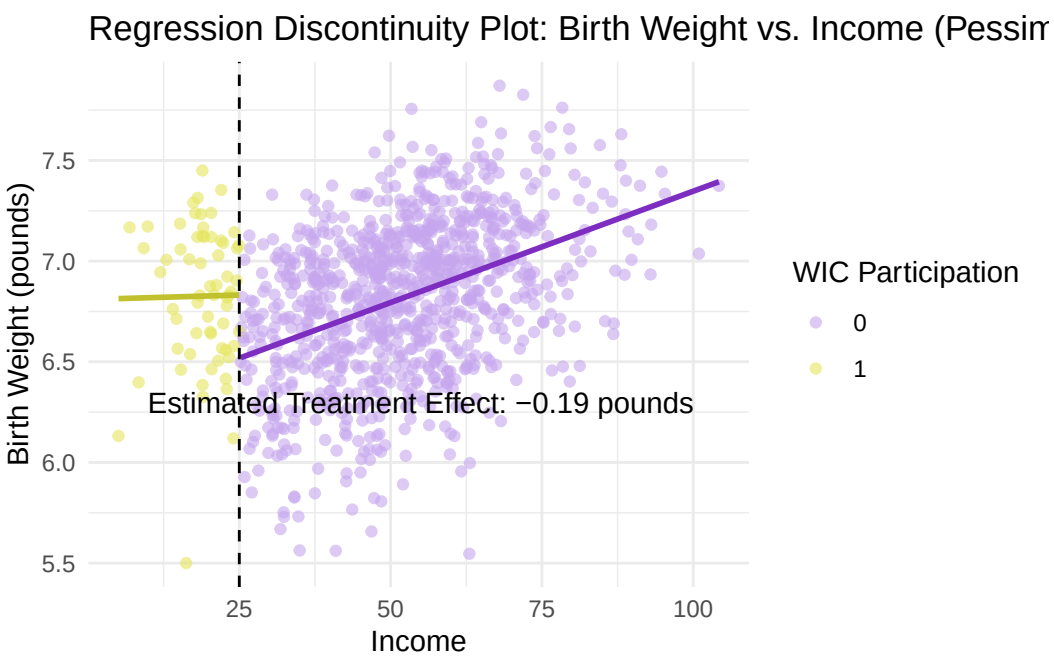
```
Sharp RD estimates using local polynomial regression.

Number of Obs.                 1000
BW type                       mserd
Kernel                   Triangular
```

```
VCE method                       NN

Number of Obs.                   63          937
Eff. Number of Obs.              31           41
Order est. (p)                    1            1
Order bias  (q)                   2            2
BW est. (h)                   5.090        5.090
BW bias (b)                   7.231        7.231
rho (h/b)                     0.704        0.704
Unique Obs.                      63          937

=============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
  Conventional   -86.392    71.872    -1.202     0.229  [-227.258 , 54.474]
        Robust         -         -    -1.143     0.253  [-263.197 , 69.256]
=============================================================================
```

```r
# Create a data frame for visualization (Pessimistic Approach)
visual_data_pessimistic <- data.frame(
  income = running_variable_pessimistic,
  birth_weight = outcome_variable_pessimistic,
  wic_participation = simulated_data_pessimistic$wic_participation_pessimistic
)

# Convert birth weight to pounds
visual_data_pessimistic$birth_weight_pounds <- visual_data_pessimistic$birth_weight / 453.592

# Extract treatment effect estimate from rdrobust
treatment_effect_pounds_pessimistic <- (rdd_results_pessimistic$Estimate[1] / 453.592)  # Co

ggplot(visual_data_pessimistic, aes(x = income, y = birth_weight_pounds, color = factor(wic_p
  geom_point(alpha = 0.6) +
  geom_smooth(data = subset(visual_data_pessimistic, income <= cutoff), method = "lm", color
  geom_smooth(data = subset(visual_data_pessimistic, income > cutoff), method = "lm", color =
  geom_vline(xintercept = cutoff, linetype = "dashed", color = "black") +
  scale_color_manual(
    values = c("0" = "#C6A6EE", "1" = "#E5E55D"),
    name = "WIC Participation"
  ) +
  labs(
    title = "Regression Discontinuity Plot: Birth Weight vs. Income (Pessimistic Approach)",
```

```
    x = "Income",
    y = "Birth Weight (pounds)"
  ) +
  annotate(
    "text",
    x = 55,
    y = max(visual_data_pessimistic$birth_weight_pounds) * 0.8,
    label = paste("Estimated Treatment Effect:", round(treatment_effect_pounds_pessimistic, 2
    size = 4,
    color = "black"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```



This regression discontinuity plot was created using the pessimistic fake data to analyze the regression discontinuity design for WIC. The estimated treatment effect of WIC participation on birth weight is 0.2 pounds, as indicated. This suggests that, even with the use of the pessimistic data, there is still an increase in birth weight for individuals who partake in the WIC program.

```
Sharp RD estimates using local polynomial regression.

Number of Obs.                 1000
BW type                      Manual
Kernel                   Triangular
VCE method                       NN

Number of Obs.                   63          937
Eff. Number of Obs.              53          120
Order est. (p)                    1            1
Order bias  (q)                   2            2
BW est. (h)                  10.000       10.000
BW bias (b)                  10.000       10.000
rho (h/b)                     1.000        1.000
Unique Obs.                      63          937

=============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
  Conventional   -78.966    53.720    -1.470     0.142  [-184.255 , 26.324]
        Robust         -         -    -0.882     0.378  [-215.612 , 81.826]
=============================================================================

Sharp RD estimates using local polynomial regression.

Number of Obs.                 1000
BW type                      Manual
Kernel                   Triangular
VCE method                       NN

Number of Obs.                   63          937
Eff. Number of Obs.              31           41
Order est. (p)                    1            1
Order bias  (q)                   2            2
BW est. (h)                   5.000        5.000
BW bias (b)                   5.000        5.000
rho (h/b)                     1.000        1.000
Unique Obs.                      63          937

=============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
```

Table 1: Treatment Effect Estimates from Different RDD Models (Grams)

Model	Estimate
RDD Optimistic (h=10)	-228.84377
RDD Optimistic (h=5)	-257.91325
RDD Pessimistic (h=10)	-78.96575
RDD Pessimistic (h=5)	-87.41419

```
  Conventional   -87.414    72.305    -1.209     0.227  [-229.130 , 54.302]
        Robust         -         -    -1.069     0.285  [-273.309 , 80.346]
==============================================================================
```

Below is the results of robustness tests for the optimistic and pessimistic approach, showing treatment effect size on birth weight (grams).

Conclusion

A pessimistic and optimistic approach was used to analyze WIC's impact using Regression Discontinuity Design (RDD), which provided robust evidence of its effectiveness in reducing adverse outcomes such as low birth weights.

- **Pessimistic Scenario:**
  - Treatment effect: -498.522 grams (95% CI: -550.915 to -446.128)
  - Highly significant (z = -18.649, $p < 0.001$)
- **Optimistic Scenario:**
  - Treatment effect: -661.954 grams (95% CI: -839.354 to -484.555)
  - Significant (z = -7.313, $p < 0.001$)

These findings justify continued or increased funding for the program and show the importance of addressing food insecurity and health disparities. Also, the evidence supports refining resource allocation and eligibility criteria to maximize program efficiency and expand its positive impact.

If no significant effect had been found, potential gaps in program implementation, such as insufficient participation rates and barriers to accessing benefits, would prompt reevaluation of WIC's components, such as nutritional offerings or service mechanisms, to address participant needs better.